

ODTE Options Strategy

Study Zero-day-to-expiration options to help trading decisions

Motivation

Research Question

Challenges

Data Source

Target Formulation

Evaluation

Black Scholes Analytic Model

Feature Engineering

Quantile Regression Model

Future Work

Motivation

Motivation

ODTE options went from a niche product to roughly half the market in just a few years.

- ODTE options have large trading volume
- Growth has outpaced research coverage

Research Question

Research Question

Given the **observable options-market state** (eg. implied volatility, greeks, open-interest exposures) at an intraday snapshot, can we estimate a **probability distribution for the underlying's closing price** at end-of-day — and does that distribution provide decision-relevant information for ODTE option positioning?

Challenges

Challenges

- Data and market microstructure
 - Intraday data is extremely noisy
 - No intraday open interest because OI is only updated once per day
 - Some greeks become explosive as time to maturity approaches 0
- Modeling challenge
 - Hard to model distribution due to non-stationarity of market regime
 - Multiple intraday snapshot but only one closing price realization per trading day
- Evaluation
 - Hard to validate against true probabilities due to only one realized closing price

Data Source

Data Source

This project uses Theta Data <https://www.thetadata.net/> Options data and focuses on SPY data. A standard option data plan is required to run this project.

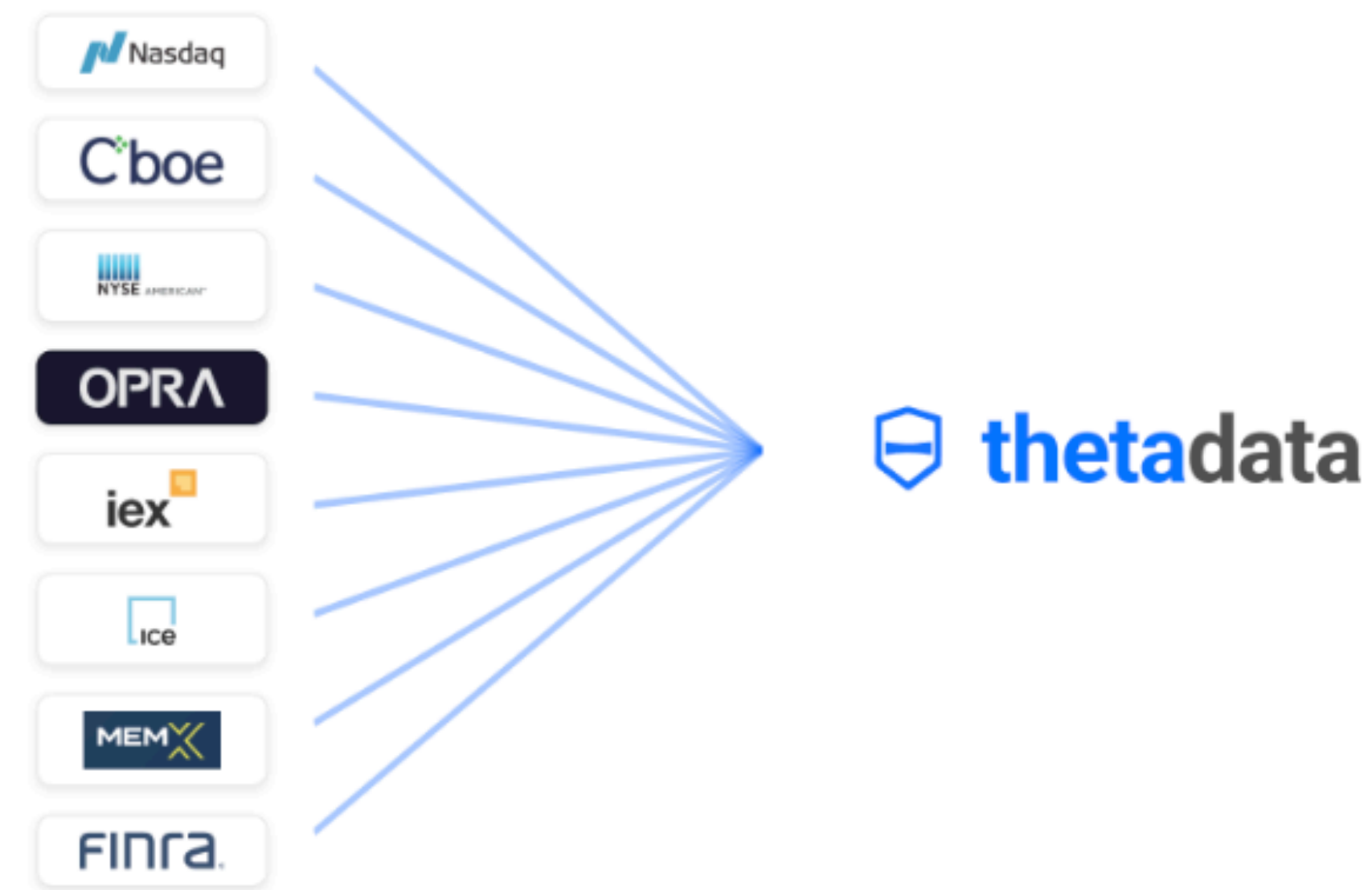
We've released a Python Library!

**Every strategy is
only as good as
its data.**

Professional-grade options, stocks, indices, and interest rates data.
Millisecond latency. Unfiltered trade and quote data. Comprehensive Greeks.

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[Documentation](#)



Data Source

First order greeks: https://docs.thetadata.us/operations_python/option_history_greeks_first_order.html

SPY ODTE Options Snapshot — 2026-06-01																
symbol	expiration	strike	right	timestamp	bid	ask	delta	theta	vega	rho	epsilon	lambda	implied_vol	iv_error	underlying_timestamp	underlying_price
SPY	2026-06-01	760.000000	CALL	2026-06-01 13:45:00-04:00	0.45	0.47	0.380	-3.566	4.635	0.074000	-0.074100	626.793600	14.300%	0.001600	2026-06-01 13:45:00-04:00	759.460000
SPY	2026-06-01	760.000000	CALL	2026-06-01 14:00:00-04:00	0.38	0.40	0.380	-3.401	4.369	0.065700	-0.065800	739.550400	12.860%	0.000000	2026-06-01 14:00:00-04:00	759.540000
SPY	2026-06-01	760.000000	CALL	2026-06-01 14:15:00-04:00	0.26	0.27	0.316	-3.142	3.817	0.047800	-0.047900	919.205900	11.910%	0.003400	2026-06-01 14:15:00-04:00	759.380000
SPY	2026-06-01	760.000000	CALL	2026-06-01 14:30:00-04:00	0.39	0.40	0.442	-3.773	3.924	0.057300	-0.057400	859.796700	11.910%	0.000400	2026-06-01 14:30:00-04:00	759.820000
SPY	2026-06-01	760.000000	CALL	2026-06-01 14:45:00-04:00	0.39	0.40	0.450	-4.412	3.592	0.048700	-0.048700	875.571100	12.690%	0.001000	2026-06-01 14:45:00-04:00	759.850000
SPY	2026-06-01	760.000000	CALL	2026-06-01 15:00:00-04:00	0.24	0.25	0.332	-4.808	2.947	0.028700	-0.028800	1050.452200	13.520%	0.001000	2026-06-01 15:00:00-04:00	759.520000
SPY	2026-06-01	760.000000	CALL	2026-06-01 15:15:00-04:00	0.21	0.22	0.307	-4.520	2.853	0.026600	-0.026600	1113.660500	13.130%	-0.001200	2026-06-01 15:15:00-04:00	759.460000
SPY	2026-06-01	760.000000	CALL	2026-06-01 15:30:00-04:00	0.24	0.25	0.369	-4.317	3.063	0.031900	-0.032000	1166.493800	11.660%	0.002000	2026-06-01 15:30:00-04:00	759.680000
SPY	2026-06-01	760.000000	CALL	2026-06-01 15:45:00-04:00	0.04	0.05	0.107	-1.755	1.491	0.009200	-0.009200	1998.726700	9.760%	0.012100	2026-06-01 15:45:00-04:00	759.010000
SPY	2026-06-01	760.000000	PUT	2026-06-01 13:45:00-04:00	1.03	1.06	-0.612	-3.754	4.662	-0.119600	0.119400	-447.494200	15.280%	-0.000900	2026-06-01 13:45:00-04:00	759.460000
SPY	2026-06-01	760.000000	PUT	2026-06-01 14:00:00-04:00	0.86	0.88	-0.615	-3.501	4.387	-0.106800	0.106600	-536.622700	13.470%	0.000400	2026-06-01 14:00:00-04:00	759.540000
SPY	2026-06-01	760.000000	PUT	2026-06-01 14:15:00-04:00	0.88	0.90	-0.679	-3.192	3.845	-0.103100	0.102900	-578.929100	12.300%	0.000300	2026-06-01 14:15:00-04:00	759.380000
SPY	2026-06-01	760.000000	PUT	2026-06-01 14:30:00-04:00	0.60	0.61	-0.554	-3.979	3.929	-0.072200	0.072100	-702.222300	12.790%	-0.000100	2026-06-01 14:30:00-04:00	759.820000
SPY	2026-06-01	760.000000	PUT	2026-06-01 14:45:00-04:00	0.55	0.56	-0.549	-4.473	3.594	-0.059500	0.059400	-757.248600	13.080%	0.000900	2026-06-01 14:45:00-04:00	759.850000
SPY	2026-06-01	760.000000	PUT	2026-06-01 15:00:00-04:00	0.72	0.73	-0.666	-4.773	2.951	-0.057800	0.057700	-703.180000	13.620%	0.000000	2026-06-01 15:00:00-04:00	759.520000
SPY	2026-06-01	760.000000	PUT	2026-06-01 15:15:00-04:00	0.75	0.76	-0.690	-4.507	2.861	-0.059900	0.059800	-698.504200	13.280%	0.001000	2026-06-01 15:15:00-04:00	759.460000
SPY	2026-06-01	760.000000	PUT	2026-06-01 15:30:00-04:00	0.56	0.57	-0.630	-4.281	3.065	-0.054600	0.054600	-853.738400	11.760%	0.000500	2026-06-01 15:30:00-04:00	759.680000
SPY	2026-06-01	760.000000	PUT	2026-06-01 15:45:00-04:00	1.01	1.03	-0.907	-1.414	1.346	-0.078700	0.078600	-675.804200	9.170%	-0.000900	2026-06-01 15:45:00-04:00	759.010000
SPY	2026-06-01	759.000000	PUT	2026-06-01 13:30:00-04:00	0.91	0.96	-0.614	-2.998	4.902	-0.133100	0.132800	-500.466000	12.930%	0.000500	2026-06-01 13:30:00-04:00	758.510000
SPY	2026-06-01	759.000000	PUT	2026-06-01 13:45:00-04:00	0.52	0.54	-0.400	-3.790	4.703	-0.078100	0.078000	-573.997800	15.230%	-0.001300	2026-06-01 13:45:00-04:00	759.460000

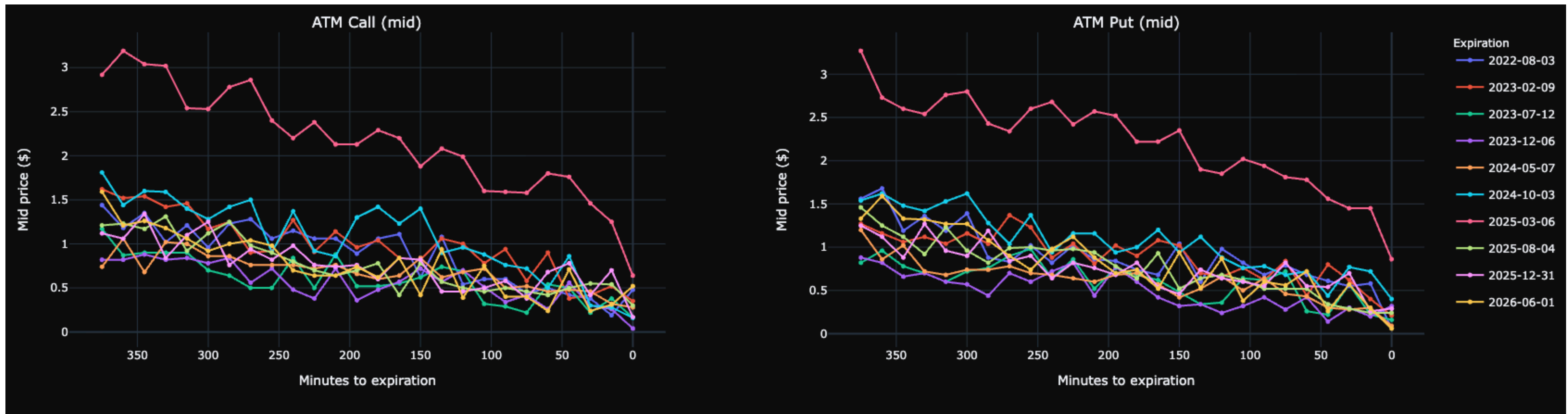
Data Source

Open interest: https://docs.thetadata.us/operations_python/option_history_open_interest.html

SPY ODTE Options Open Interest — 2026-06-01						
symbol	expiration	strike	right	timestamp	open_interest	
SPY	2026-06-01	505.000000	CALL	2026-06-01 06:30:02.444000-04:00	0	
SPY	2026-06-01	910.000000	PUT	2026-06-01 06:30:02-04:00	0	
SPY	2026-06-01	910.000000	CALL	2026-06-01 06:30:02-04:00	0	
SPY	2026-06-01	505.000000	PUT	2026-06-01 06:30:01.019000-04:00	0	
SPY	2026-06-01	718.000000	PUT	2026-06-01 06:30:01.177000-04:00	5070	
SPY	2026-06-01	782.000000	CALL	2026-06-01 06:30:02.444000-04:00	714	
SPY	2026-06-01	718.000000	CALL	2026-06-01 06:30:00.920000-04:00	31	
SPY	2026-06-01	782.000000	PUT	2026-06-01 06:30:01.481000-04:00	0	
SPY	2026-06-01	739.000000	PUT	2026-06-01 06:30:02.130000-04:00	2090	
SPY	2026-06-01	675.000000	PUT	2026-06-01 06:30:03.330000-04:00	679	
SPY	2026-06-01	675.000000	CALL	2026-06-01 06:30:01.361000-04:00	7	
SPY	2026-06-01	739.000000	CALL	2026-06-01 06:30:01.361000-04:00	879	
SPY	2026-06-01	696.000000	PUT	2026-06-01 06:30:03.320000-04:00	190	
SPY	2026-06-01	760.000000	CALL	2026-06-01 06:30:02.827000-04:00	16080	
SPY	2026-06-01	696.000000	CALL	2026-06-01 06:30:02.860000-04:00	2	
SPY	2026-06-01	760.000000	PUT	2026-06-01 06:30:03.066000-04:00	1471	
SPY	2026-06-01	570.000000	CALL	2026-06-01 06:30:01.481000-04:00	0	
SPY	2026-06-01	570.000000	PUT	2026-06-01 06:30:02.130000-04:00	50	
SPY	2026-06-01	717.000000	PUT	2026-06-01 06:30:02.827000-04:00	727	
SPY	2026-06-01	781.000000	CALL	2026-06-01 06:30:02.860000-04:00	1328	

Data Source

Option mid price vs. time to expiration



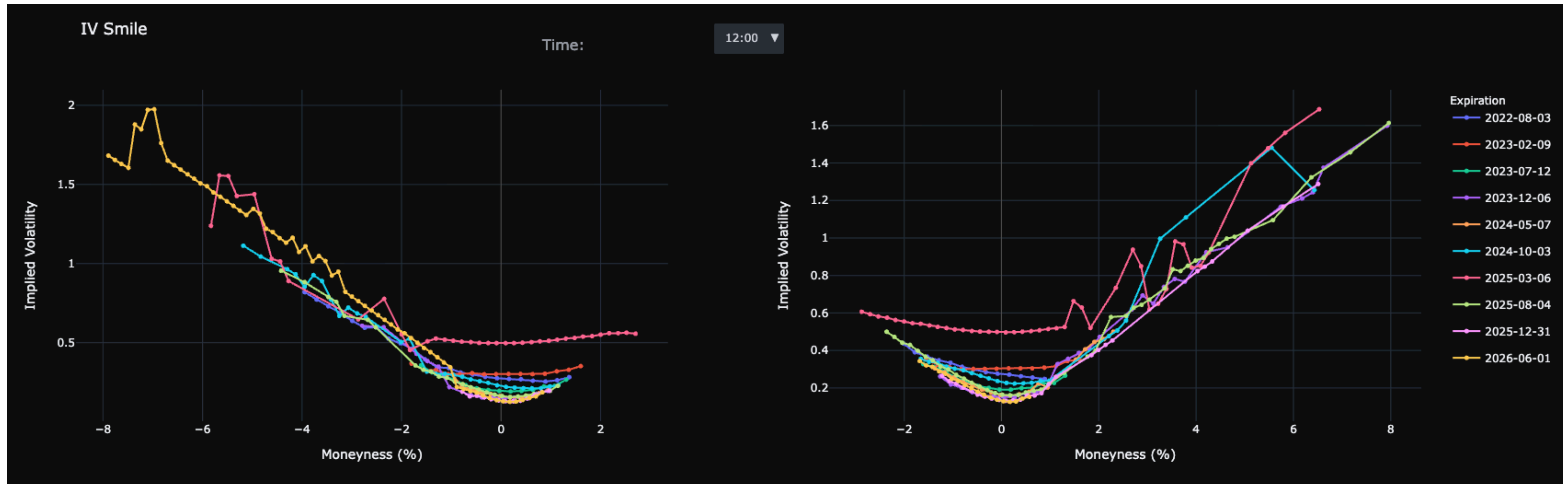
Data Source

SPY Underlying Price



Data Source

IV Smile: Implied Volatility vs. moneyness



Target Formulation

Target Formulation

Closing prices are extracted from raw greeks underlying_price column at the closing time 16:00, and similarly for opening price at 9:30.

Closing price



$$\text{Log return} = \ln \left(\frac{\text{closing price}}{\text{price at snapshot}} \right)$$

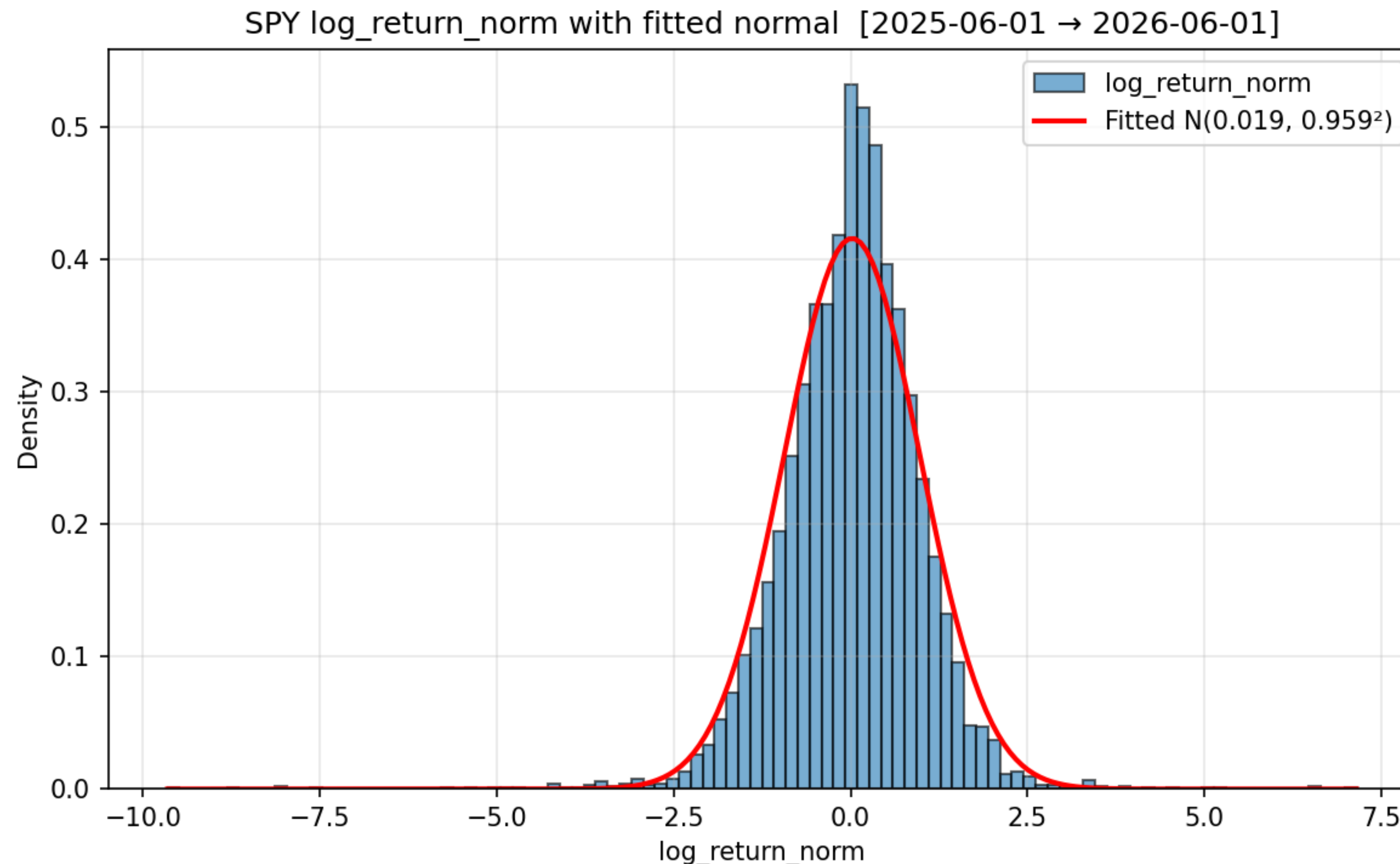


$$\text{Log return norm} = \frac{\text{log return}}{\text{atm iv} \cdot \sqrt{\text{ttm in years}}}$$

Norm factor in denominator chosen based on Black Scholes model

Target Formulation

SPY log return norm histogram with fitted normal distribution:



Evaluation

Evaluation: Pinball Loss

For a quantile τ , predicted value $\hat{y}$ and true value y , the pinball loss is:

$$L_{\tau}(y, \hat{y}) = \begin{cases} \tau(y - \hat{y}), & \text{if } y \geq \hat{y} \\ (1 - \tau)(\hat{y} - y), & \text{if } y < \hat{y} \end{cases}$$

- Naturally penalizes asymmetrically, good for distribution prediction
 - eg. For 90th percentile (90% chance price end up below here) prediction, under-predicting is penalized 9x more than over-predicting
- No assumption on distributional shape

Black Scholes *Analytic* Model

Black Scholes Analytic Model

Black Scholes Model assumes a geometric Brownian motion for the price movement, so the **prediction target log return norm** can be calculated from risk-free interest rate μ , time to maturity in year, and volatility σ , which follows a normal distribution:

$$N\left(\frac{\mu\sqrt{ttm}}{\sigma} - \frac{1}{2}\sigma\sqrt{ttm}, 1\right)$$

Derivation

t : current time

T : closing time

S_t : underlying price at time t

Z_t : Geometric Brownian motion with volatility σ

Let $y \sim N(0,1)$:

$$(Z_T - Z_t) \sim N(0, T-t) = \sqrt{T-t} y$$

Derived from Ito's Lemma:

$$S_T = S_t e^{(\mu - \frac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t} y}$$

log return $\ln\left(\frac{S_T}{S_t}\right) = (\mu - \frac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t} y$

log return norm $\frac{\ln\left(\frac{S_T}{S_t}\right)}{\sigma\sqrt{T-t}} = \frac{\mu\sqrt{T-t}}{\sigma} - \frac{1}{2}\sigma\sqrt{T-t} + y$

Baseline: μ risk-free interest rate (around 4%)

$T-t$ time to maturity in year

σ atm IV

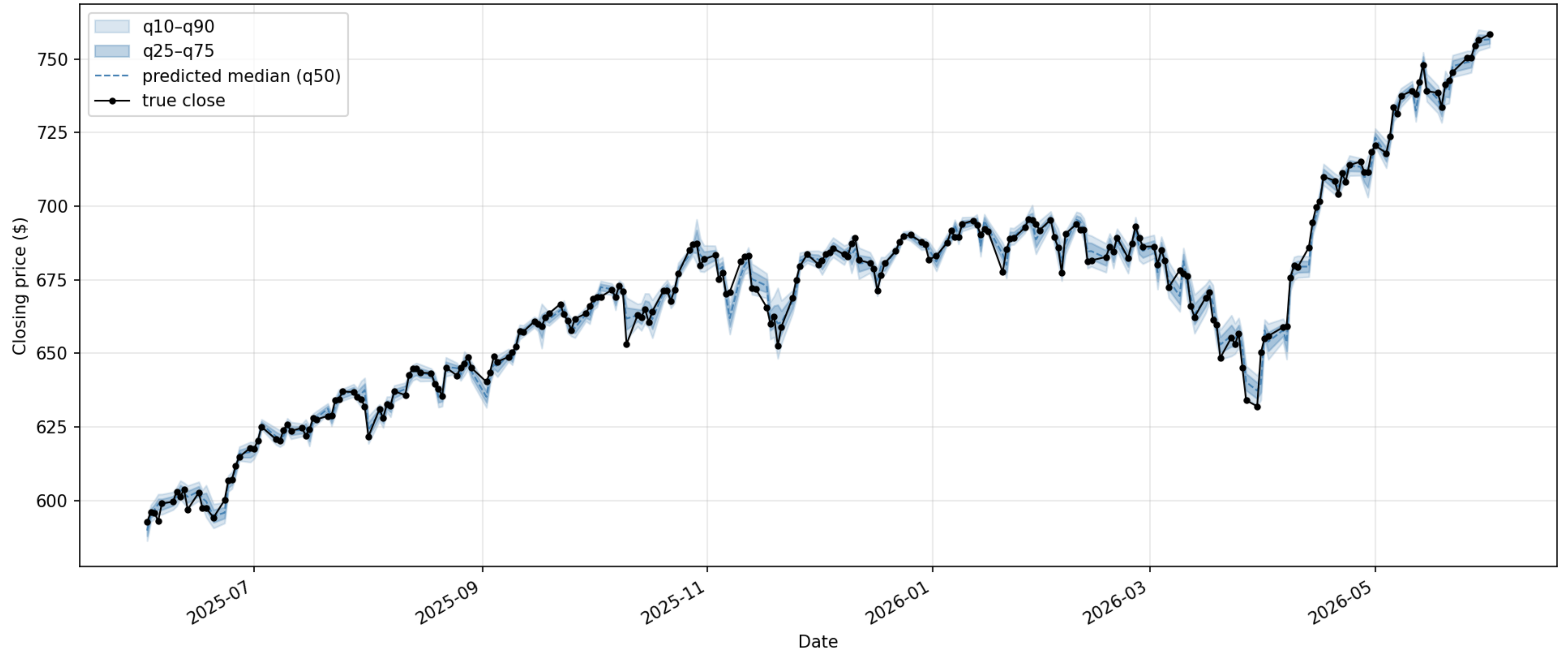
$$N\left(\frac{\mu\sqrt{ttm}}{IV} - \frac{1}{2} IV\sqrt{ttm}, 1\right)$$

Result (SPY 2025-6-1 to 2026-6-1)

```
Model 1 (Black-Scholes) - SPY [2025-06-01 -> 2026-06-01]
6274 rows (251 days)
Pinball loss (normalized space):
  q=0.10  pinball=0.1769
  q=0.25  pinball=0.2988
  q=0.50  pinball=0.3532
  q=0.75  pinball=0.2797
  q=0.90  pinball=0.1613
  avg     pinball=0.2540
```

Result (SPY 2025-6-1 to 2026-6-1)

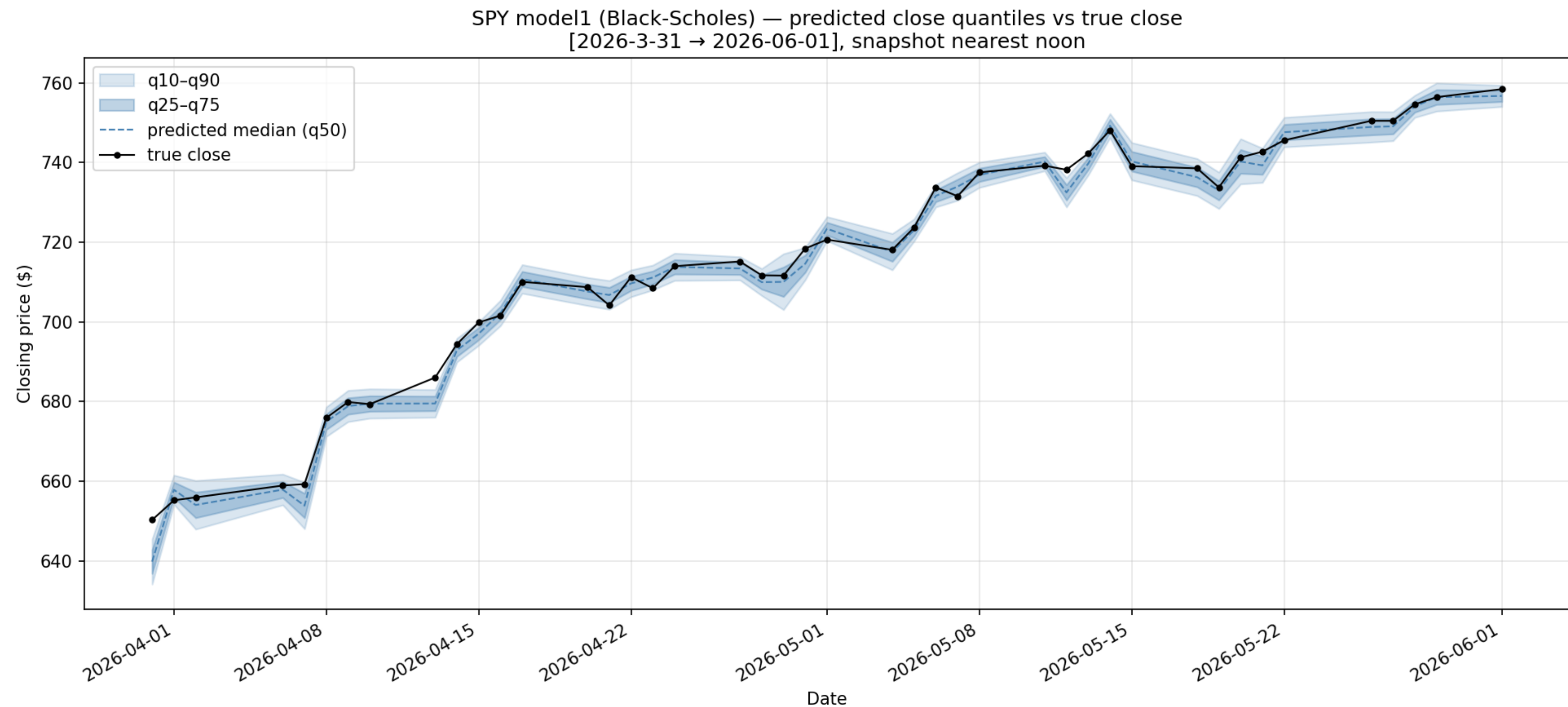
SPY model1 (Black-Scholes) — predicted close quantiles vs true close
[2025-01-01 → 2026-06-01], snapshot nearest noon



Limitation

Black Scholes model assumes geometric Brownian motion.

As a result, this model is not good with noticeable trends.



Feature Engineering

From Raw Data

symbol, expiration, strike, right, timestamp, bid, ask, delta, theta, vega, rho, epsilon, lambda, implied_vol, iv_error, underlying_timestamp, underlying_price

`log_return_from_open` calculated from $\ln(\text{underlying price} / \text{opening price})$

Black Scholes features `d1` and `gamma` calculated from Black Scholes

Open Interest Related

- `open_interest` is read from raw OI data
- delta exposure `dex`, gamma exposure `gex` and theta exposure `tex` are calculated from OI and the corresponding greeks
- `total_oi` is the put + call OI at a (day, strike)
- `put_oi_fraction` = put OI / total OI at a (day, strike)
- `max_oi_strike` is the strike price with max OI at a day
- `oi_concentration_top3` is (top 3 largest put + call OI) / total OI in all strikes on the day
- `distance_to_max_oi` is underlying price - (max OI strike on that day)

Bid-Ask Features

- $\text{bid_ask_mid} = (\text{bid} + \text{ask}) / 2$
- $\text{bid_ask_spread} = \text{ask} - \text{bid}$
- $\text{bid_ask_spread_norm} = \text{bid ask spread} / \text{bid ask mid}$

Aggregate: Greeks

Exposure:

- `net_dex`, `net_gex` and `net_tex` are aggregated from DEX, GEX and TEX
- `net_gex_norm`, `net_dex_norm`, `net_tex_norm` are normalized using underlying price

`theta_decay` = `net_tex_norm` / ttm in hours

Aggregate: Implied Volatility

`atm_iv` at-the-money IV, the IV where the strike is closest to the underlying price

`iv_call_25d` and `iv_put_25d` are the call and put IV where delta is closest to ± 0.25

`iv_skew_25d` = `iv_put_25d` - `iv_call_25d`

`iv_smile_curvature_25d` = `iv_put_25d` + `iv_call_25d` - 2 * `atm_iv`

Quantile Regression Model

Quantile Regression

This model has 5 quantiles: 10th (10% chance price ends up below here), 25th, 50th, 75th, 90th.

Quantile regression is chosen because no assumption about the distribution is required, thus skews, fat tails or other unexpected behavior can be naturally captured.

Training loss is pinball loss.

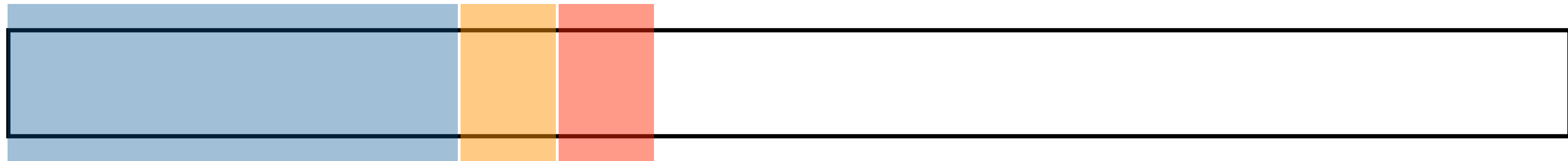
LightGBM

- Natively supports pinball loss through built-in quantile objective.
- Fast to train and iterate.
- Has sensible regularization defaults to avoid overfitting.
- Some greys explode as time to maturity approaches 0, and LightGBM handles exploding features well.

Walk Forward (Rolling-Origin)

Intuition: markets are non-stationary, and we want to fit on the most recent data to predict the period right after it.

train val test

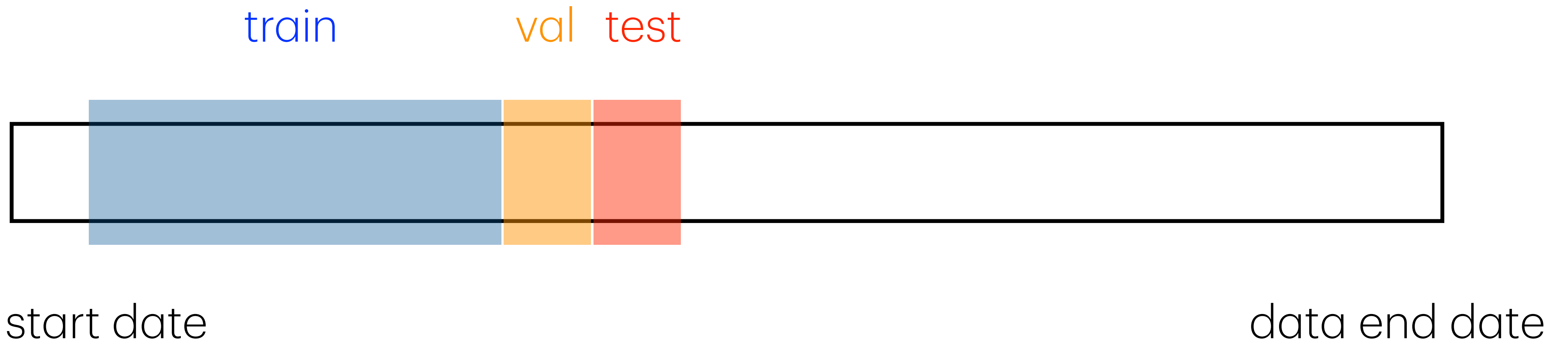


data start date

data end date

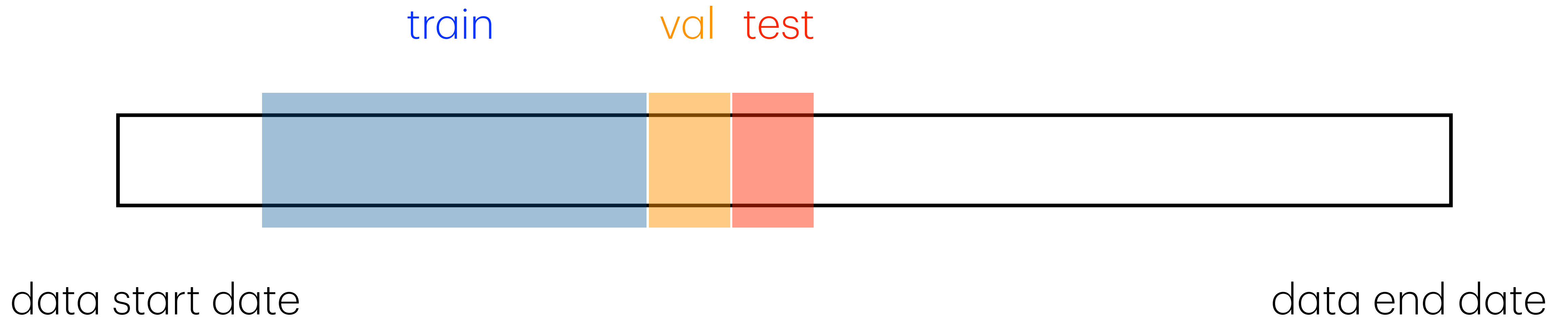
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Walk Forward (Rolling-Origin)

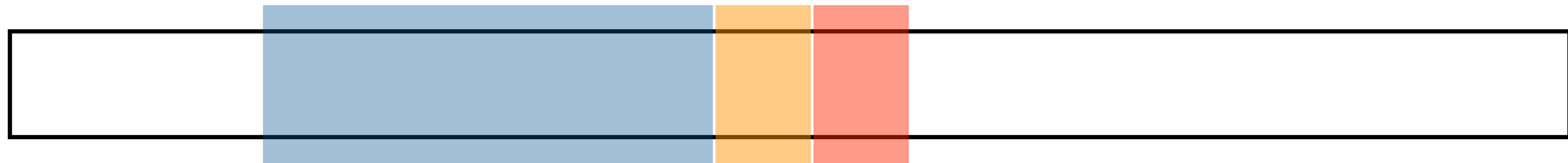
Intuition: markets are non-stationary, and we want to fit on the most recent data to predict the period right after it.



Walk Forward (Rolling-Origin)

Intuition: markets are non-stationary, and we want to fit on the most recent data to predict the period right after it.

train val test



data start date

data end date

Result

Better than Black Scholes analytic model in recent data where there's a noticeable rising trend in both pinball and coverage.

```
Mean pinball (normalized) -> model: 0.21053   Black-Scholes: 0.21398   (model +1.6% vs BS)
Mean pinball (log-return) -> model: 6.850e-04   Black-Scholes: 7.280e-04
```

Per-quantile (lower pinball is better; coverage should approx the quantile level):

	quantile	model_pinball_norm	bs_pinball_norm	model_coverage	bs_coverage	model_beats_bs
0	0.100000	0.15173	0.15159	0.151	0.046	False
1	0.250000	0.25076	0.26200	0.286	0.154	True
2	0.500000	0.28386	0.29392	0.535	0.415	True
3	0.750000	0.23655	0.22804	0.794	0.794	False
4	0.900000	0.12974	0.13436	0.926	0.960	True

Future Work

Future Work

- Keep refining current quantile regression model.
- Work on a deeper feature engineering and data construction.
- Broaden evaluation beyond statistical loss.
- Evaluate model on other underlying.
- Build an application layer to actually help trading decision making.

ODTE Options Strategy

Study Zero-day-to-expiration options to help trading decisions